

Costing your own service

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Cost **one episode of care** that your department delivers, from the hospital's side, **bottom up**: count what is used, then price it. Rough numbers are fine. The point is to find out which lines matter and which numbers you do not actually know.

1 DEFINE THE EPISODE

Department _____ Names in the group _____

The episode _____

Where it starts and ends, e.g. "one elective laparoscopic cholecystectomy, admission to discharge" or "one new-patient OPD visit for suspected TB, up to diagnosis"

Perspective: ☐ hospital (provider) ☐ health system ☐ abridged societal (adds what the patient pays)

Cost year: 20____

2 MAP THE PATHWAY

One box per stage the patient passes through. Everything you count below should belong to a box.

stage 1	stage 2	stage 3	stage 4	stage 5
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3 STAFF TIME

Cost per minute = monthly gross salary ÷ minutes actually available per month (working days × productive hours × 60; leave, teaching and admin time are not available)

Cadre	Monthly gross (₹)	Minutes available / month	₹ per minute	Minutes on this episode	Cost (₹)
Staff nurse (constructed)	60,000	$22 \times 6 \times 60 = 7,920$	7.58	45	341
A Staff subtotal					

4 DRUGS, CONSUMABLES AND INVESTIGATIONS

Item	Quantity	Unit price (₹)	Price from	Cost (₹)
IV cannula (constructed)	2	35	hospital store rate	70
B Drugs, consumables and investigations subtotal				

Worked rows in grey are constructed to show the arithmetic. They are not real salaries or prices.

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5 EQUIPMENT: ANNUITISE, DO NOT CHARGE THE PURCHASE PRICE

Equivalent annual cost = purchase price ÷ annuity factor

Cost per use = equivalent annual cost ÷ uses per year

Equipment	Price (₹)	Life (yr)	Factor	Uses / yr	Cost (₹)
Ultrasound (constructed)	5,00,000	10	8.530	2,000	29
C Equipment subtotal					

Example: $5,00,000 \div 8.530 = ₹58,617$ a year; $\div 2,000$ scans = ₹29 a scan. Machine shared with other departments? Use its total uses per year.

Useful life (years)	Annuity factor at 3%
5	4.580
7	6.230
8	7.020
10	8.530
15	11.938
20	14.877
30 (building)	19.600

Factor = $[1 - (1.03)^{-n}] \div 0.03$. The 3% matches the Indian Reference Case discount rate.

6 SHARED OVERHEADS: THE COSTS NOBODY OWNS

Administration, laundry, kitchen, electricity, water, security, CSSD, the building itself. Pick one method and say which.

Step-down share = annual cost of the shared service × your department's share of the allocation base (beds, floor area, staff or patient-days) ÷ episodes your department delivers per year

Shared service	Annual cost (₹)	Allocation base and your share	Episodes / yr	Cost (₹)
D Overhead subtotal (or a stated mark-up: ____ % of A + B, source: _____)				

7 ADD IT UP, AND COMPARE

Cost of one episode to the hospital = A + B + C + D	₹
PM-JAY package rate for the nearest matching procedure (if one exists)	₹
National Health System Cost Database figure (if one exists)	₹

8 THREE QUESTIONS TO ANSWER BEFORE YOU REPORT BACK

(a) Which line was biggest? Did it surprise you?

(b) Which single number are you least sure of? How much would the total move if it were half, or double?

(c) What does the patient pay that is not on this sheet? Think of drugs bought outside, tests sent out, travel, food, an attendant, days of lost work.

Question (b) is a one-way sensitivity analysis done by hand. Question (c) is the difference between the provider perspective and the abridged societal perspective the Indian Reference Case asks for.